Event-TimeRAF: Event-Aware Retrieval-Augmented Foundation Models for Explainable Air Quality Forecasting Under Concept Drift

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Abstract—**Urban air-quality forecasting is important for public-health warnings, environmental management, and city-scale decision support. Recent time-series foundation models have shown useful zero-shot forecasting capability, and TimeRAF enriches such models by retrieving relevant time-series sequences from a knowledge base and incorporating them into a frozen time-series foundation model. However, air-pollution variation is often influenced by outside context, including meteorology, holidays, traffic activity, wildfire smoke, and other event-driven factors. In this draft, we propose Event-TimeRAF, an event-aware retrieval-augmented forecasting framework for Los Angeles County PM2.5 prediction. The proposed methodology combines historical air-quality windows, weather features, calendar context, retrieved time-series patterns, source-recorded environmental events, distribution-shift indicators, and evidence-based explanation generation. The manuscript presents the motivation, related work, problem formulation, methodology, and experimental plan; no empirical performance result is claimed before the official-source pipeline is executed and audited.**

Index Terms—**Air quality forecasting, PM2.5, retrieval-augmented forecasting, time-series foundation models, TimeRAF, concept drift, explainable forecasting.**

I. Introduction

Air pollution is a major urban risk factor, and fine particulate matter (PM2.5) is especially important because it is associated with respiratory and cardiovascular health effects [1]. Forecasting PM2.5 can support public-health alerts, exposure reduction, transport planning, and environmental policy. In Los Angeles County, air quality can change because of meteorological conditions, traffic patterns, seasonal behavior, industrial activity, wildfire smoke, and unusual environmental events. These properties make air-quality forecasting a difficult time-series forecasting task.

Classical forecasting models and machine-learning models such as ARIMA, tree-based regressors, recurrent neural networks, and Transformer-based models can learn historical patterns when training and testing conditions are similar. However, urban air quality is strongly context-dependent. A model trained only on past numerical observations may perform poorly during unusual weather, public holidays, traffic disruptions, fire incidents, dust events, or other external shocks. This limitation is closely connected to concept drift, where the data-generating process changes over time.

Time-series foundation models (TSFMs) have recently been proposed to improve generalization across forecasting tasks. Models such as TimesFM, Moirai, MOMENT, Chronos, and Tiny Time Mixers (TTM) aim to transfer forecasting knowledge learned from large time-series corpora to new downstream domains [2]–[6]. TimeRAF takes this direction further by introducing retrieval augmentation for zero-shot time-series forecasting [7]. It builds a time-series knowledge base, retrieves relevant candidate sequences, and integrates the retrieved knowledge using Channel Prompting while keeping the TSFM backbone frozen.

TimeRAF is a strong base method, but the retrieved knowledge is mainly numerical time-series data. For air-quality forecasting, external context is important: rainfall can wash out particulates, low wind speed can increase stagnation, holidays can change traffic patterns, and wildfire or smoke events can produce abrupt changes. Therefore, a natural extension is to retrieve not only similar time-series patterns, but also event, weather, and calendar context relevant to the forecast horizon.

This draft proposes Event-TimeRAF, an event-aware retrieval-augmented framework for explainable Los Angeles County PM2.5 forecasting under distribution shift. The planned contributions are:

- an event-aware retrieval-augmented forecasting framework for urban PM2.5 prediction;
- a hybrid knowledge-base design that combines air-quality windows, weather summaries, calendar information, and event/news records;
- a TimeRAF-style retrieval baseline using random retrieval and similarity-based retrieval over historical windows;
- simple concept-drift indicators based on distribution shift, retrieval similarity, weather changes, and event bursts;
- evidence-based explanation generation using retrieved historical cases, event context, weather drivers, drift indicators, and feature importance;
- a planned ablation study comparing no retrieval, random retrieval, time-series retrieval, event retrieval, drift-aware retrieval, and the full Event-TimeRAF framework.

The Results and Discussion section contains only the reporting plan and placeholders until the official data pipeline and model experiments have been executed. No experimental performance values are invented.

II. Literature Review and Related Work

A. Time-Series Forecasting

Time-series forecasting has traditionally been studied through statistical models, machine-learning regressors, and

This is an implementation-stage draft prepared for academic supervision. Experimental results are intentionally left as placeholders until the full official-source evaluation is completed.

deep neural networks. Autoregressive and seasonal models remain useful baselines because they are interpretable and efficient. Gradient-boosted tree models such as XGBoost and LightGBM are widely used for tabular forecasting with lagged and rolling features [8], [9]. Recurrent models such as LSTM and GRU can capture sequential dependence [10], [11]. More recently, Transformer-based forecasting models have used patching and channel-independent designs to handle long lookback windows more efficiently [12]. For air-quality forecasting, these approaches are helpful, but they may still be sensitive to distribution shifts and unobserved external events.

B. Time-Series Foundation Models

TSFMs aim to provide transferable forecasting ability across datasets and domains. TimesFM uses a decoder-only architecture for time-series forecasting [2]. Moirai proposes a universal forecasting Transformer trained on a large-scale open time-series archive [3]. MOMENT provides a family of open time-series foundation models for general-purpose time-series tasks [4]. Chronos converts time-series values into tokens and trains language-model architectures for probabilistic forecasting [5]. TTM proposes compact pre-trained models for efficient zero/few-shot forecasting [6]. These models motivate the use of pre-trained forecasting backbones, but practical domain adaptation is still difficult when the downstream series responds strongly to local events and environmental conditions.

C. Retrieval-Augmented Forecasting

Retrieval-augmented generation first became popular in NLP by combining parametric models with non-parametric memory [13]. Dense Passage Retrieval (DPR) introduced dense dual-encoder retrieval for open-domain question answering [14]. TimeRAF adapts the retrieval-augmentation idea to zero-shot time-series forecasting [7]. It constructs a time-series knowledge base, retrieves top-k candidate windows using a learnable retriever, and combines them through Channel Prompting. The base paper reports TTM-Base as the main backbone, context length 512, forecast horizon 96 in most experiments, and $k = 8$ as the default number of retrieved candidates. Its ablations compare random retrieval, cosine retrieval, learnable retrieval, alternative integration mechanisms, candidate counts, knowledge-base choices, and backbone choices.

Event-TimeRAF follows the TimeRAF principle but changes both the knowledge source and the target domain. Instead of relying only on numerical time-series retrieval, it proposes a hybrid knowledge base containing historical PM2.5 windows, weather summaries, calendar context, and event/news evidence. The first implementation will be TimeRAF-inspired rather than a full reproduction: random retrieval and cosine retrieval will be implemented first, while a dual-encoder MLP retriever and Channel-Prompting-style neural fusion remain advanced extensions.

D. Air-Quality Forecasting and Data Sources

PM2.5 forecasting often uses historical pollutant measurements together with meteorological factors such as temperature, humidity, wind speed, precipitation, and pressure. The US Environmental Protection Agency publishes quality-controlled hourly observations through AirData and the Air Quality System [15]. NOAA's Integrated Surface Database provides hourly meteorological observations from surface stations [16]. NOAA also publishes structured Storm Events records and historical Hazard Mapping System fire and smoke products [17], [18]. These official sources support a source-preserving Los Angeles County forecasting pipeline without requiring manually authored event records.

E. Explainable and Event-Aware Forecasting

Explainability is important in environmental forecasting because users need to understand why a model predicts a rise or fall in pollution. SHAP values provide a model-agnostic framework for estimating feature contributions [19]. However, explanations based only on feature importance may miss temporal and contextual cues. Event-TimeRAF therefore plans to generate explanations from structured evidence: recent PM2.5 persistence, weather flags, retrieved similar historical windows, retrieved event records, calendar context, and drift indicators. The explanation module will use templates grounded in stored evidence, not unrestricted language-model generation.

III. Problem Formulation

Let x_t denote the PM2.5 measurement at time t. For a forecast time t, the historical air-quality lookback window is

$$X_t = [x_{t-L+1}, x_{t-L+2}, \ldots, x_t], \tag{1}$$

where L is the lookback length. Let W_t denote weather covariates aligned with the lookback window, C_t denote calendar features, and E_t denote event context retrieved from an event/news knowledge base. The target is the future PM2.5 sequence

$$Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \ldots, x_{t+H}], \tag{2}$$

where H is the forecasting horizon.

The forecasting goal is to learn a function

$$\hat{Y}_{t+1:t+H} = F_\theta(X_t, W_t, C_t, R_t^{ts}, R_t^{ev}, D_t), \tag{3}$$

where R_t^{ts} is retrieved time-series knowledge, R_t^{ev} is retrieved event knowledge, and D_t contains concept-drift indicators. In the initial implementation, the first target setting is $L = 168$ hours and $H = 24$ hours. Longer horizons of 48, 72, and 96 hours are planned after the 24-hour pipeline is stable.

The planned evaluation uses a time-based split:

$$\mathcal{T}_{train} < \mathcal{T}_{val} < \mathcal{T}_{test}, \tag{4}$$

where the earliest 70% of observations are used for training, the next 15% for validation, and the latest 15% for testing. Random splitting is not used because it would leak future temporal information into training.

Table I summarizes the essential notation.

TABLE I
MAIN SYMBOLS USED IN THE PROPOSED FORMULATION

Symbol	Meaning
L	Lookback window length
H	Forecast horizon
X_t	Historical PM2.5 sequence at time t
W_t	Weather covariates
C_t	Calendar/context features
R_t^{ts}	Retrieved time-series knowledge
R_t^{ev}	Retrieved event/news knowledge
D_t	Drift indicators
$\hat{Y}_{t+1:t+H}$	Predicted future PM2.5 values

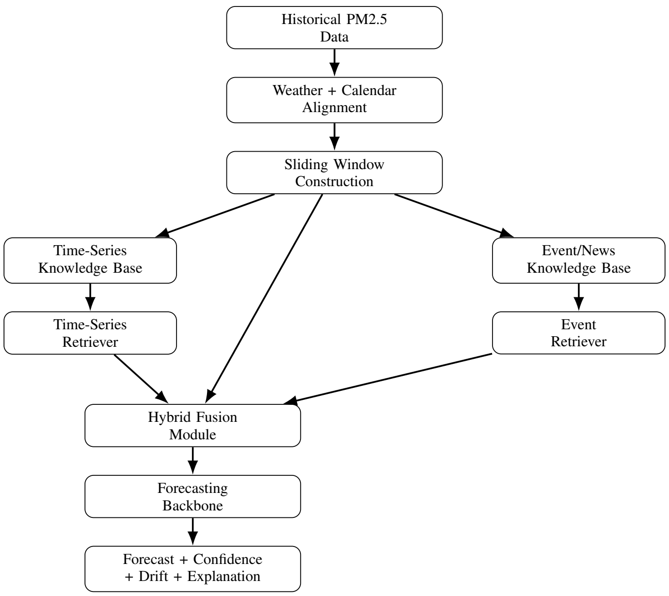

Fig. 1. Planned Event-TimeRAF architecture. The figure is conceptual and does not report experimental results.

IV. METHODOLOGY

A. Overview

Fig. 1 shows the planned Event-TimeRAF pipeline. The first step is to align air-quality observations with weather, calendar, and availability-filtered event features. The system then creates sliding windows and a time-series knowledge base. In the MVP, one hybrid retriever ranks historical windows using PM2.5, weather, calendar, and event-context similarity. A separate source-record lookup links recent event identifiers to explanations; it does not supply future event labels to the model. The retrieved evidence is combined with structured features to generate a forecast, uncertainty proxy, drift score, and explanation.

B. Relationship to TimeRAF

TimeRAF retrieves top-k candidate time-series windows from a curated knowledge base and incorporates the retrieved candidates using Channel Prompting [7]. It uses a learnable dual-encoder retriever, dot-product similarity, a frozen TSFM backbone, and candidate augmentation during retriever training. Event-TimeRAF keeps the retrieval-augmentation principle, but applies it to Los Angeles County PM2.5 forecasting and expands the knowledge sources beyond numerical sequences.

The first implementation will not be a complete clone of TimeRAF. Instead, it will implement a TimeRAF-style retrieval baseline using random retrieval and cosine similarity. A dual-encoder MLP retriever, candidate augmentation, and TSFM-based Channel Prompting are reserved for later extensions after the Kaggle-compatible MVP is reproducible.

C. Dataset Construction

Hourly PM2.5 observations will be taken from US EPA AirData/AQS parameter 88101 for Los Angeles County [15]. The data audit will select one monitoring site using coverage and continuity criteria. Timestamps will be converted to `America/Los_Angeles` with explicit daylight-saving handling. Short input gaps may be filled only from past-available observations; targets will never be interpolated. Weather covariates will come from the nearest suitable NOAA Integrated Surface Database station [16]. Calendar features will include hour, weekday, weekend, month, season, US federal holidays, and California holidays.

All normalization statistics will be fitted on the training split only. The same normalization procedure will be used for query windows and retrieved candidate windows, following the TimeRAF requirement that input and retrieved sequences use consistent preprocessing.

D. Time-Series Knowledge Base

The time-series knowledge base will contain historical PM2.5 windows. Each record will store a window identifier, start time, end time, input window values, future window values, weather summary, calendar summary, embedding vector or normalized window vector, data split, and normalization identifier. Knowledge-base origins are separated by 192 hours, equal to the 168-hour input and 24-hour future combined. For every query, a candidate is eligible only when its complete future ends before the query input begins. Validation and test retrieval is additionally restricted to training-history windows. These rules exclude self-matches, near-duplicate windows, and overlap with the query context.

Given a query window X_t, the time-series retriever returns

$$R_t^{ts} = \{r_{t,1}^{ts}, r_{t,2}^{ts}, \ldots, r_{t,k}^{ts}\}, \tag{5}$$

where k is the number of retrieved candidates. The default planned setting is $k = 8$, with sensitivity analysis for $k \in \{1, 4, 8, 16\}$ and optionally $k = 32$.

E. Event Knowledge Base

The event knowledge base will contain source-preserving records relevant to Los Angeles County. NOAA Storm Events is the required structured source, while NOAA Hazard Mapping System fire and smoke records are an optional enrichment [17], [18]. Planned fields include event identifier, event start and end, information-availability timestamp, location, category, source, source URL, title, and summary. Event categories include wildfire, smoke, excessive heat, high wind, heavy rain, flood, dust, traffic, industrial activity, and policy restrictions.

For the first implementation, model inputs represent events with hourly publication counts, rolling 24-hour, 72-hour, and 168-hour counts, category-specific 72-hour counts, active-event counts, and an event-burst ratio. Recent source event identifiers are retained only as explanation evidence. Every event used as a model input must be available by the forecast origin under a recorded availability rule. Evaluation distinguishes events published in the prior 72 hours, events active at the origin, and recorded event intervals overlapping the target horizon. Target overlap is used only as a post-hoc label and never as a model input. NOAA Storm Events does not provide a machine-readable publication timestamp; consequently, the implementation records event start as a retrospective availability assumption. Results that use this assumption will be labeled sensitivity experiments rather than strict operational forecasts. Strict experiments require source records with genuine publication or issuance timestamps.

F. Hybrid Retrieval Score

The planned hybrid retrieval score combines time-series similarity, weather relevance, calendar relevance, and event relevance:

$$s(q,i) = \alpha s_{ts}(q,i) + \beta s_w(q,i) + \gamma s_c(q,i) + \delta s_e(q,i), \quad (6)$$

where s_{ts} is time-series similarity, s_w is weather similarity, s_c is calendar similarity, and s_e is event relevance. Initial weights are planned as $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$, and $\delta = 0.2$. These weights will only be adjusted with validation data after implementation.

G. Fusion and Forecasting Backbone

The MVP fusion method combines PM2.5 lag/rolling features, weather features, calendar features, retrieved time-series summary features, event features, and drift indicators. The forecasting backbone uses 24 direct XGBoost regressors, one for each forecast horizon, because this design is efficient on Kaggle and provides a strong tabular baseline. A later extension may use an MLP residual fusion module inspired by Channel Prompting:

$$z_i = \mathrm{Concat}(h_t, r_{t,i}), \quad (7)$$

$$h_t^* = h_t + \frac{1}{k}\sum_{i=1}^{k} \mathrm{MLP}(z_i), \quad (8)$$

where h_t is the input representation and $r_{t,i}$ is the representation of the i-th retrieved candidate. This mirrors the TimeRAF idea of preserving the original input representation while adding uniformly averaged information extracted from retrieved candidates.

H. Concept Drift Detection

The implementation represents operational distribution shift using transparent indicators rather than claiming perfect concept-drift identification. The indicators are recent mean shift, recent variance shift, retrieval similarity drop, weather-pattern shift, and event-burst ratio. A simple drift score can be defined as

$$D_t = \lambda_1 z_{\mu,t} + \lambda_2 z_{\sigma,t} + \lambda_3(1-\bar{s}_{ts,t}) + \lambda_4 z_{w,t} + \lambda_5 b_{event,t}, \quad (9)$$

where $z_{\mu,t}$ and $z_{\sigma,t}$ are normalized mean and variance shifts, $\bar{s}_{ts,t}$ is average retrieval similarity, $z_{w,t}$ is weather-pattern shift, and $b_{event,t}$ is an event-burst score. The implemented components are robustly scaled against training data and averaged with equal weight; the alert threshold is selected from validation scores.

TABLE II
METHODOLOGICAL COMPARISON BETWEEN TIMERAF AND EVENT-TIMERAF

Aspect	TimeRAF	Event-TimeRAF Draft
Target domain	General zero-shot TSF	Los Angeles County PM2.5 forecasting
Knowledge base	Time-series windows	PM2.5 windows, weather, calendar, events
Retriever	Learnable dual encoder	Random/cosine first; learnable later
Integration	Channel Prompting	Feature fusion first; Channel-Prompting-inspired later
Backbone	Frozen TSFM	Direct XGBoost MVP; frozen TSFM gate
Explanations	Retrieved-case visualization	Evidence-based forecast explanation
Drift handling	Not central	Explicit drift indicators

I. Explanation Generation

The explanation module will generate evidence-based explanations. It will use signed local XGBoost contributions averaged across the 24 direct horizon models, weather flags, retrieved historical windows, retrieved event records, calendar context, forecast direction, component-level drift evidence, and an uncertainty proxy combining retrieved-trajectory spread with validation residual error. It will not generate explanations from an unrestricted language model. Each explanation must be traceable to stored machine-readable evidence.

Table II summarizes the methodological relation between TimeRAF and Event-TimeRAF.

V. EXPERIMENTAL SETUP

This section describes the planned experimental setup. The experiments have not yet been performed.

A. Planned Data

Table III lists the planned data sources and feature groups. The initial study period is 2019–2024, subject to site-level PM2.5 coverage and event-data audit results.

TABLE III
PLANNED DATA SOURCES AND FEATURE GROUPS

Group	Planned Source	Example Features
Air quality	US EPA AirData/AQS	hourly PM2.5, site, timestamp
Weather	NOAA Global Hourly/ISD	temperature, humidity, rain, wind, pressure
Calendar	Generated US/California features	hour, weekday, weekend, season, holidays
Events	NOAA Storm Events; optional HMS	event counts, categories, source records

B. Forecasting Task

The initial task predicts the complete PM2.5 sequence from one to 24 hours ahead using a 168-hour lookback window. The planned split uses the earliest 70% of observations for training, the next 15% for validation, and the latest 15% for testing. Random splitting will not be used.

C. Baselines

The planned baselines are:

- persistence baseline;
- daily and weekly seasonal naive baselines;
- direct XGBoost models using PM2.5 lag/rolling features;
- direct XGBoost models using PM2.5, weather, and calendar features;
- random-retrieval baseline;
- cosine TimeRAF-inspired retrieval baseline;
- Event-TimeRAF without drift indicators;
- full Event-TimeRAF with event features and drift indicators.

The NOAA Storm Events variants are retrospective event-availability sensitivity experiments because event start is used when a publication timestamp is unavailable. They will not be presented as strict real-time event forecasts.

A frozen Chronos-Bolt model and its retrieval-augmented forecast-level fusion are the planned foundation-model publication gate. Other models such as ARIMA/SARIMA, LSTM/GRU, PatchTST, or TTM may be added if time and runtime allow.

D. Metrics

The planned metrics are mean squared error (MSE), mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), symmetric MAPE (sMAPE), and R^2. MSE is included for comparability with the TimeRAF base paper, while MAE and RMSE are important for interpretation in air-quality forecasting. Separate target-event, drift-period, and normal-period errors will be reported only when a subset contains at least 50 unique forecast origins. Smaller subsets will be described by counts and cases without inferential comparison.

E. Ablation Plan

Table IV gives the planned ablation matrix. The entries describe planned model variants only; no result is reported.

Additional planned analyses include candidate-count sensitivity for $k \in \{1, 4, 8, 16\}$, optional $k = 32$, uniform versus similarity-weighted retrieval aggregation, and knowledge-base size sensitivity if runtime allows.

VI. RESULTS AND DISCUSSION

At this draft stage, the Kaggle-oriented data, retrieval, forecasting, drift, explanation, and evaluation modules have been implemented, but the complete 2019–2024 official-source experiment has not been executed and frozen. Therefore, this section intentionally does not report numerical values, performance plots, or comparative claims. A one-year engineering smoke test is not treated as an experimental result.

After implementation, this section will report:

- main forecasting results across all baselines and Event-TimeRAF variants;
- ablation results for weather, calendar, time-series retrieval, event retrieval, and drift indicators;
- retrieval sensitivity results for different values of k;
- drift-period versus normal-period error;
- event-period case studies, explicitly separated into retrospective-sensitivity and strict-availability settings;
- example evidence-based explanations;
- computational cost or inference-time comparison if feasible.

The discussion will be written only after real outputs are generated. In particular, the final paper will not claim that Event-TimeRAF improves over baselines unless the saved evaluation tables support that claim.

VII. CONCLUSION

This draft proposed Event-TimeRAF, an event-aware retrieval-augmented framework for explainable Los Angeles County PM2.5 forecasting under distribution shift. The proposed framework extends the TimeRAF idea by combining time-series retrieval with weather, calendar, source-recorded environmental events, drift indicators, and evidence-based explanation generation. The implementation is designed as a Kaggle-compatible pipeline using official US data, direct XGBoost baselines, leakage-safe retrieval, and a frozen-TSFM publication gate.

Because the complete experiment has not yet been executed, this manuscript does not report performance claims. The next step is to run and audit the full 2019–2024 pipeline on Kaggle, complete the frozen-TSFM publication gate, freeze validation choices, evaluate the test set once, and populate the paper only from saved artifacts. The conclusion will be revised after those results are available.

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TABLE IV
PLANNED ABLATION STUDY

Model	PM2.5	Weather	Calendar	TS Retrieval	Event Retrieval	Drift	Explanation
Persistence	Yes	No	No	No	No	No	No
Seasonal naive	Yes	No	Yes	No	No	No	No
XGBoost-basic	Yes	No	No	No	No	No	No
XGBoost-weather-calendar	Yes	Yes	Yes	No	No	No	No
Random-retrieval baseline	Yes	Yes	Yes	Random	No	No	Limited
Cosine TimeRAF-inspired	Yes	Yes	Yes	Cosine	No	No	Limited
Event-TimeRAF-no-drift	Yes	Yes	Yes	Yes	Yes	No	Yes
Event-TimeRAF-full	Yes	Yes	Yes	Yes	Yes	Yes	Yes

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